# Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

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***Abstract*—Melanoma classifiers pick up artifacts such as dark corners, rulers and ink, and a prototype network's explanation ("this part looks like that prototype") can point at the artifact. On ISIC 2019, dark corners and bright structures are associated with melanoma even after stratifying by source. Existing remedies supervise accuracy or activations; in dermoscopy, none controls *which* training patches prototypes may become. We propose lesion-anchored prototype learning with two components: a lesion-attraction objective on the normalized spatial attention of each prototype, and a projection step that admits only cells that are at least half lesion and nearly free of detected artifacts, so every prototype comes from a lesion-dominated patch by construction, although its receptive field extends beyond it. The masks need no new annotation (Dice 0.94 on held-out expert masks), and inference needs none. In a same-backbone, leakage-audited comparison with lesion-grouped statistics, the model raises the AUC of ProtoPNet from 0.825 to 0.865 (0.859 without an auxiliary artifact-suppression term), not significantly different from the strongest mask-supervised ProtoPNet (0.867). Through lesion attraction, it roughly halves the decision-flip rate of ProtoPNet when artifacts are inserted around bit-identical lesion pixels (dark corners: 10.9%→5.2%) and flips fewer decisions than any mask-loss baseline for dark corners and ink, with more stable explanations than all of them. The projection costs no accuracy and keeps all prototypes on lesion-dominated cells, which no mask loss does. Adding the original mask loss further weakens the real dark-corner shortcut; that loss alone still generalizes better to Derm7pt. Code and logs: https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

## I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], but they also learn *shortcuts* [2], features that correlate with the label in the training data without being related to the disease. In dermoscopy, surgical skin markings raise the melanoma probability of a deep classifier [3], models perform far above chance when the lesion is removed [4], [5], and dark corners, rulers, gel bubbles and ink are frequent and unevenly distributed across sites [6]–[8].

For clinical use, Rudin [9] argues for models that are interpretable by design. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so that a decision reads "this part looks like that part". Nothing, however, stops a prototype from being a piece of a ruler or a black corner; the shortcut then survives and is even presented as a reasonable explanation [11], and such prototypes have been observed in dermoscopy [12], [13]. **Research gap.** Prior work leaves three things open (Table I). (i) Debiasing methods for dermoscopy [7], [14], [15] improve robustness to artifacts, but their models are black boxes. (ii) For prototype networks, mask supervision penalizes activation outside annotated regions [12], [16], and human feedback removes or switches off unwanted prototypes [13], [17]; none of these methods constrains the *push* step, which decides which training patch a prototype finally *is*, and feedback acts only after a user has found the problem. (iii) Artifact insertion has tested the *decisions* of classifiers [13], [15], [18], but whether the *explanations* of a prototype network stay on the lesion when artifacts are added or removed has, to our knowledge, not been tested. **Contributions.** We fill these gaps as follows.

- **Lesion attraction on the normalized prototype attention.** An objective on the spatial attention of every prototype pulls it towards lesion tissue without shrinking the activations. It needs no new annotation and drives the gains in accuracy and decision robustness: removing it lowers the test AUC from 0.865 to 0.830 ($p < 0.001$) and roughly doubles the flip rates under inserted dark corners, rulers and ink. The main model gains 4.0 AUC points over ProtoPNet (3.4 with the two core components alone), not significantly different from the strongest mask-supervised ProtoPNet (the formulation of the closest prior work [12]), whose explanations rely three times more on artifacts.
- **Lesion- and artifact-constrained projection.** The push step admits only latent cells whose input patch is at least half lesion and at most 5% detected artifact, so every prototype comes from such a cell *by construction*, a guarantee that loss-based supervision does not give: no tested weight or formulation of the mask loss puts all 20 prototypes on such cells in any run (on average 9–18). On six independently trained networks, each projected both ways, this step changes neither the AUC nor, consistently,

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the flip rates or the explanation maps, and thresholds from 0.3 to 0.9 behave alike: it fixes what an explanation shows, not how the model decides. A foreground-restricted projection appears concurrently as an appendix variant for natural images [19]; new here are the artifact exclusion, the combination with lesion attraction and the test against shortcuts.

- **A counterfactual test of decisions *and* explanations.** We insert four artifact types and remove real hair and ruler marks while every lesion pixel stays bit-identical, with an inpainting-only null control and a complementary test on unedited images, and measure how far both the decision and the prototype explanation move.
- **A leakage-controlled, mask-audited evaluation.** We group near-duplicates in addition to lesion identifiers, validate every mask source and the artifact detectors, compare eight configurations, four further variants of the mask-loss baseline and its combination with our model, all with the same backbone and three seeds, use lesion-grouped statistics, and test on three external sets.

An auxiliary artifact-suppression term, kept in the main model, brought no significant gain on its own; we report it as a largely negative result, not as a contribution.

## II. Related Work

**Artifacts and shortcuts in dermoscopy.** Bissoto *et al.* annotated seven artifact types, found that simple debiasing does not remove their bias [5] and built artifact-based domain-generalization benchmarks [6]. Removing dark corners changes accuracy only marginally but moves activations onto the lesion [8], and pasted colored patches make a CNN call malignant lesions benign [18]. EPVT [7] learns domain prompts for artifact groups, Bevan and Atapour-Abarghouei [14] unlearn artifact information adversarially, and Reveal-to-Revise (R2R) [15] finds band-aids, rulers and skin markers semi-automatically, corrects the model with RRR, CDEP or ClArC and tests it on images with pasted artifacts. All of these models are black boxes.

**Prototype networks and explanation supervision.** The similarity that a prototype network suggests can be misleading [11], which motivates testing explanations rather than taking them at face value [20]. RRR [21] penalizes input gradients inside annotated irrelevant regions, and CDEP [22] contextual-decomposition importance, demonstrated on ISIC images with colored patches. IAIA-BL [16] adds a fine-annotation loss that penalizes prototype activation outside regions marked by radiologists. Closest to us, Correia *et al.* [12] apply this loss to melanoma-versus-nevus classification with lesion masks from a segmentation network; our PPN-FA baselines follow this approach, in its original formulation and in a weaker one (Section VI-G). ProtoPDebug [17] and the medical study of PIP-Net [13] let a user mark confounded prototypes, which are then forgotten or switched off. None of these methods constrains the projection step, and none tests prototype explanations under artifact interventions. Concurrently, Lin and Wright [19] study architectural prototype diversity on birds and faces with a hinge loss on the background share of each prototype's activation and, in an appendix, a projection restricted to patches with more than 50% foreground; we target shortcut artifacts, exclude them in the projection, need no new mask annotation and test with counterfactuals.

TABLE I

Positioning against related work. Annot.: human input; Interp.: interpretable by design; Where: constrains where the model looks; Prov.: constrains which training patches prototypes become; CF: artifact interventions testing decisions (D) and explanations (E). Lin and Wright [19] study prototype diversity on birds and faces, not artifacts ([†]appendix variant).

| Method | Annot. | Interp. | Where | Prov. | CF |
|---|---|---|---|---|---|
| EPVT [7], Bevan [14] | image labels | – | – | – | – |
| R2R [15] | semi-auto. | – | ✓ | – | D, E |
| RRR [21], CDEP [22] | pixel masks | – | ✓ | – | – |
| ProtoPNet [10] | none | ✓ | – | – | – |
| IAIA-BL [16] | expert masks | ✓ | ✓ | – | – |
| Correia *et al.* [12] | auto. masks | ✓ | ✓ | – | – |
| ProtoPDebug [17] | user | ✓ | ✓ | post hoc | – |
| PIP-Net (med.) [13] | user | ✓ | – | post hoc | D |
| Lin, Wright [19] | object masks | ✓ | ✓ | foreground[†] | – |
| **Ours** | **none new** | ✓ | ✓ | **les., no art.** | **D, E** |

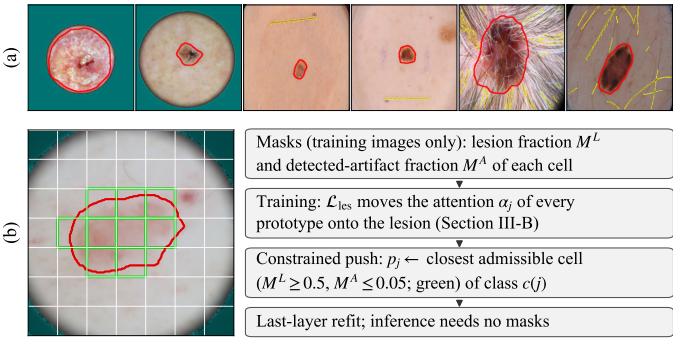



Fig. 1. (a) Automatic masks on ISIC 2019 images (red: lesion; cyan: dark corners; yellow: rulers, third and fourth, and hair, last two); artifact masks never overlap the lesion. (b) Method overview: the $7 \times 7$ cells of a test image that the constrained projection would admit (green), and the pipeline.

## III. Method

### A. Prototypical part network

We build on a DenseNet-121 backbone [23] (overview: Fig. 1(b)). Two $1 \times 1$ convolutions with a final sigmoid turn its output into a latent map $z \in [0,1]^{H \times W \times D}$, here with $H = W = 7$ and $D = 128$. The network keeps $m = 20$ prototypes $p_j \in \mathbb{R}^D$, ten for each class, and $c(j)$ denotes the class of prototype $j$. For every cell we compute the distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and convert it into a similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_c \mathcal{L}_{\text{clst}} - \lambda_s \mathcal{L}_{\text{sep}} + \lambda_l \mathcal{L}_{\text{les}} + \lambda_a \mathcal{L}_{\text{art}}, \quad (1)$$

where $\mathcal{L}_{\text{CE}}$ is a class-weighted cross-entropy and the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10] ($\lambda_c = 0.8$,

$\lambda_s = 0.08$ as originally). Our model combines the two core components, lesion attraction $\mathcal{L}_{\text{les}}$ ($\lambda_l = 1$) and the constrained projection of Section III-C, with an auxiliary artifact-suppression term $\mathcal{L}_{\text{art}}$ ($\lambda_a = 1$) whose ablation (Section VI-H) shows small, mixed effects. All use the masks of Section IV.

### B. Lesion attraction (core component)

The masks undergo the same geometric augmentation as the image; average pooling to the $7 \times 7$ grid gives, for each cell, the fraction covered by lesion, $M^L(h, w)$, and by artifacts, $M^A(h, w)$. Where a prototype looks is described by its spatial attention $\alpha_j = \text{softmax}_{h,w}(s_j/\tau)$, $\tau = 0.5$, a soft version of the max-pooling location that produces $a_j$. Each prototype is weighted by $\omega_j = a_j / \sum_k a_k$ (treated as a constant), so that the prototypes the model actually uses count most. With $L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$, the attention mass of prototype $j$ on the lesion of image $i$,

$$\mathcal{L}_{\text{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \qquad (2)$$

where $\tilde{\omega}$ renormalizes $\omega$ over the prototypes of the true class $y_i$, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the batch images whose crop still contains lesion. The logarithm penalizes prototypes that almost ignore the lesion much more than those that already look at it, and since $\alpha_j$ sums to one, raising the lesion mass lowers the mass on everything else, artifacts included. Unlike the fine-annotation loss of IAIA-BL [12], [16], which penalizes the magnitude of activation outside a mask, $\mathcal{L}_{\text{les}}$ acts on the normalized attention and leaves the scale of the activations to the classification losses. Unlike the concurrent hinge on the background share of the activation [19], it rewards the lesion mass logarithmically and weights each prototype by its use.

### C. Lesion- and artifact-constrained projection (core component)

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of its class; nothing prevents this patch from lying on a ruler, a black corner or healthy skin, even if the attention was well placed during training. We therefore admit only cells with $M^L \geq 0.5$ and $M^A \leq 0.05$ (Fig. 1(b)). Every prototype is then, by construction, the latent vector of a cell whose $32 \times 32$ input patch is at least half lesion and contains at most 5% detected artifact, so the "that part" of each explanation points to a patch dominated by lesion tissue. The constraint fixes where a prototype comes from, not every pixel that shapes its latent vector, since a cell also sees its surroundings (quantified in Section VI-C). After the push, only $\mathbf{W}$ is trained, with backbone, add-on layers and batch-normalization statistics fixed, so each prototype stays identical to its source patch. As five last-layer epochs did not solve this convex problem for *any* prototype model, we re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an $\ell_1$ penalty of $10^{-4}$ on negative-class connections [10]; 3,000 full-batch Adam steps, cosine-annealed learning rate from 0.02), identically for every prototype model and without validation data, and also report results without it.

## IV. Masks: Provenance, Reliability and Leakage Control

### A. Provenance and validation of the lesion masks

Every image is tagged with its mask source. (i) *Expert masks*: for the 10,015 HAM10000 images (39.5% of ISIC 2019) we use the public lesion segmentations of HAM10000 [24] unchanged. (ii) *Predicted masks*: the 15,316 BCN20000 [25] and MSK [26] images have no public masks, so we train a DeepLabV3 [27] network with an ImageNet-initialized MobileNetV3-Large [28] encoder at $256 \times 256$ (12 epochs, AdamW, one-cycle learning rate up to $5 \times 10^{-4}$, binary cross-entropy plus soft Dice, random flips and $90^\circ$ rotations) *only* on the 8,163 HAM10000 images of our training split, which is fixed beforehand. Its prediction is averaged with that of the flipped image, thresholded at 0.5, reduced to the largest connected component and hole-filled. Masks covering less than 0.5% of the image (19 images, 0.08%) are marked invalid and excluded from mask-based losses and metrics. On the 1,852 HAM10000 images of the validation and test splits, unseen by the segmentation network, the predicted masks reach a mean Dice of 0.9445 (median 0.9672, mean Jaccard 0.9015) against the expert masks, and on the 200 expert-segmented PH2 images [29] from another clinic and device, without adaptation, a Dice of 0.918. Section VI-I tests whether our conclusions depend on mask quality.

### B. Artifact masks

Artifact masks come from fixed, label-free image operations at $256 \times 256$ and always exclude the dilated lesion. *Dark corners* (DC) are dark regions (HSV value $< 50$) touching the border and at least 8 px thick on average; *thin dark structures* (TS: hair, ruler ticks) are found with a $13 \times 13$ black-hat filter as in DullRazor [30] plus an elongation test; *bright structures* (SP: specular highlights, gel-bubble rims, also light hairs) with an $11 \times 11$ white top-hat filter; and *ink or colored patches* (INK) as strongly saturated blue-to-violet pixels (Fig. 1(a)). An image is flagged once a mask covers more than 1%, 0.3%, 0.05% or 0.1% of it. *Validation.* Against the public manual labels of Bissoto *et al.* [5], available for 2,593 of our MSK images, the flags are precise but conservative (precision/recall: DC vs. dark corner 0.92/0.10, TS vs. hair or ruler 0.96/0.62, SP vs. gel bubble or border 0.86/0.35, INK vs. ink or patches 0.74/0.25); a visual check of 24 random DC and 24 random TS flags agrees (24 and 20 correct). Many artifacts, above all faint dark corners (37% manual vs. 4% detected prevalence), are missed, so "without artifacts" means "without *detected* artifacts": an undetected artifact can enter a prototype cell, artifact shares count detected artifacts only, and the reference group of Section VI-E contains undetected ones, which pulls $A$ toward 0.5 for every model. In the 2,092 training images with manual image-level labels, only 1.9% of the admissible cells lie on the outer ring, where dark corners lie, of an image with an undetected one (loose upper bounds for hair or rulers,

gel and ink, which the labels do not localize: 40%, 43%, 8%); none of the 5 prototypes of Ours from them is such a cell.

### C. How masks are used, and why they cannot leak

Masks enter at exactly four places: the training losses, the constrained projection, the attribution metrics and the placement of counterfactual edits. The classifier never receives a mask, so its test results do not depend on test masks. Leakage is prevented at every stage. (i) *Split first:* the split is computed from image and lesion identifiers and image content only, before any model, including the segmentation network, is trained. (ii) *Group-level split:* images are grouped by lesion identifier and by near-duplicate pairs (Section V-A), which can only remove leakage; an audit finds no group, lesion identifier or image shared between sets, and all 342 near-identical pairs (thumbnail correlation $\geq 0.97$) and all other verified near-duplicate pairs lie within one set. (iii) *Training-only supervision:* the segmentation network uses the 8,163 training-split masks only, and the 1,852 expert masks of validation and test serve only for evaluation; mask-based losses and projection use training images only, thresholds and checkpoints are chosen on validation, and the last-layer re-optimization uses training activations only. (iv) *Label-free masks:* neither the segmentation network nor the artifact detectors see a diagnosis, so masks cannot carry label information. (v) *Untouched external sets:* Derm7pt, PAD-UFES-20 and PH2 are used only for evaluation; removing external images flagged as possible near-duplicates of ISIC training images (8, 6 and 7) changes no external AUC by more than 0.004.

## V. Experimental Setup

### A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [24], BCN20000 [25], MSK [26]; 4,522 melanomas, 17.9%), resized to $256 \times 256$. Unlike common practice, we keep the 2,084 MSK images without lesion identifier. To avoid leakage we group images by lesion identifier and by near-duplicate pairs (perceptual-hash distance $\leq 8$ and thumbnail correlation $\geq 0.9$ for full and central image); the 13,259 groups are split into ten stratified folds, one for testing and one for validation (about 81/10/9% per source), with almost identical composition (Table II). External test sets are Derm7pt [31], with a dermoscopic and a clinical image for each of 1,011 cases (melanomas include *in situ* lesions and metastases), and the smartphone images of PAD-UFES-20 [32]; PH2 has no diagnoses in our copy and only checks the segmentation.

### B. Models and training

All configurations share backbone (ImageNet-pretrained DenseNet-121), input size ($224 \times 224$), data, split and schedule. **CNN**: backbone, global average pooling, linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [16] on our lesion masks as in Correia *et al.* [12], with their $\lambda = 10^{-3}$ but computed on the $7 \times 7$ maps and averaged over the batch, which weakens it (original formulations and larger weights: Section VI-G). **Ours**: lesion attraction, constrained projection and the auxiliary $\mathcal{L}_{\mathrm{art}}$ ($\lambda_l = \lambda_a = 1$ in Eq. (1)); the two core components alone are Ours-noArt. **Ablations**: without $\mathcal{L}_{\mathrm{les}}$ (Ours-noLes), without $\mathcal{L}_{\mathrm{art}}$ (Ours-noArt, the two core components only), with the standard projection (Ours-stdPush).

We train with Adam, batch size 64 and mixed precision: 5 epochs with frozen backbone (add-on layers and prototypes at $3 \times 10^{-3}$), 15 epochs of all layers (backbone $10^{-4}$, weight decay $10^{-3}$, learning rate divided by 10 every 5 epochs), then the projection and 5 last-layer epochs; the CNN is trained for the same 20 epochs. Class weights are 0.608 and 2.81. The CNN keeps its best validation checkpoint; prototype models are always evaluated after the projection, with the last-layer epoch chosen on validation. Each configuration runs with seeds 0, 1 and 2 (about 30 minutes per run on an NVIDIA T4). *Resolutions.* Images and masks are stored at $256 \times 256$ and receive the same random crop and flips before bilinear resampling to the $224 \times 224$ input (test: plain resize); pooled to the $7 \times 7$ grid, each cell covers a $32 \times 32$ input patch, and attribution maps are compared with masks at $224 \times 224$. All loss weights were fixed in advance and not tuned on test data.

### C. Evaluation protocol

**Classification.** Each run's threshold is fixed on ISIC validation (Youden index) and reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are mean

TABLE II
The ISIC 2019 split and the prevalence of detected artifacts. Lesion masks: expert for HAM10000, predicted for BCN20000 and MSK (valid for 99.91, 99.96 and 100% of the training, validation and test images). External sets: Derm7pt, 1,011 cases with a dermoscopic and a clinical image (252 melanomas); PAD-UFES-20, 2,298 images (52 melanomas); PH2, 200 images (segmentation only).

| | Train | Val. | Test | All |
|---|---|---|---|---|
| Images (% mel.) | 20,552 (17.8) | 2,459 (19.3) | 2,320 (16.8) | 25,331 (17.9) |
| *Diagnosis* | | | | |
| Melanoma | 3,657 | 475 | 390 | 4,522 |
| Nevus | 10,523 | 1,208 | 1,144 | 12,875 |
| Basal cell carc. | 2,671 | 322 | 330 | 3,323 |
| Actinic keratosis | 693 | 80 | 94 | 867 |
| Benign keratosis | 2,095 | 278 | 251 | 2,624 |
| Dermatofibroma | 193 | 16 | 30 | 239 |
| Vascular lesion | 201 | 27 | 25 | 253 |
| Squamous cell carc. | 519 | 53 | 56 | 628 |
| *Source (images / melanomas)* | | | | |
| BCN20000 | 10,037/2,299 | 1,225/310 | 1,151/248 | 12,413/2,857 |
| HAM10000 | 8,163/917 | 952/103 | 900/93 | 10,015/1,113 |
| MSK | 2,352/441 | 282/62 | 269/49 | 2,903/552 |
| *Artifact prevalence (%)* | | | | |
| Dark corners | 25.5 | 25.7 | 25.1 | – |
| Thin dark structures | 24.2 | 25.7 | 23.4 | – |
| Bright structures | 13.8 | 14.4 | 13.3 | – |
| Ink / patches | 1.6 | 1.9 | 2.0 | – |

$\pm$ std over seeds; tests use seed ensembles (mean probability of the three seeds).

**Attribution.** For a prototype model, the similarity maps of the three most activated prototypes are upsampled to $224\times224$, and their top 5% of pixels (the 95th-percentile region that ProtoPNet visualizes) form the hot region; for the CNN we use the class-activation map, which with a global-average-pooling head equals Grad-CAM [33] up to a positive factor [34]. We report the share of the hot region on the lesion and on artifacts, whether its maximum falls on an artifact (pointing), and the *enrichment*, artifact share over artifact area (above 1: more attention than the area explains).

**Counterfactuals.** Each test image is modified in four ways outside a 4-pixel margin around the lesion: a ruler along one border, 12–29 curved hairs, a dark vignette, or 2–4 purple ink arcs. Conversely, detected thin dark structures (real hair and ruler marks, 544 images) are erased by Telea inpainting [35] outside a 2-pixel margin; as a null control, a region of the same shape is shifted onto artifact-free skin and inpainted. Every lesion pixel is verified to stay bit-identical. We record the mean $|\Delta p|$ of the melanoma probability and the rate of flipped decisions and, for the three prototypes most active on the original image, the Spearman correlation $\rho$ of the top-1 map before and after and the *capture*, the part of the new hot region on inserted pixels.

**Statistics.** The 2,320 test images form 1,234 groups of the split (a lesion with its near-duplicates; 471 with several images), so we resample whole groups: ensemble AUCs get 95% CIs from 1,000 group resamples and are compared by a paired bootstrap with 2,000 (two-sided $p$), and per-image metrics, averaged over seeds and then within groups, by the paired Wilcoxon test. Derm7pt (one image per case) uses the paired DeLong test [36], [37], PAD-UFES-20 a bootstrap over lesions. The 16 tests behind the main claims (Ours vs. each of PPN-std, PPN, PPN-FA and the strongest mask-loss variant of Section VI-G on AUC, artifact share and flip rate under inserted dark corners and rulers) form the primary family, corrected with Holm's method; all other tests are secondary, uncorrected and exploratory. Image-level tests, which ignore the grouping, agree unless noted. All tests treat the trained models as fixed and capture the uncertainty of the test sample; variation over training runs is shown by the std over the three seeds and checked at the seed level in Sections VI-D and VI-F.

## VI. Results

We compare five main models (CNN, PPN-std, PPN, PPN-FA, Ours) unless stated otherwise; ablations, mask-loss variants and the artifact-suppression term follow in Sections VI-F–VI-H. Bold: best of the four main prototype models.

### A. Artifacts are associated with melanoma

In the training split, melanomas show dark corners and bright structures significantly more often than other lesions (Table III). Source could explain this (dark corners: 47% of BCN vs. 5% of HAM images, and BCN has the highest melanoma rate), but stratifying by source does not remove it: the Mantel–Haenszel odds ratios stay above 1, and for dark corners the OR exceeds 1 in every source (BCN 1.41, HAM 3.10, MSK 2.24), for bright structures in BCN and HAM (2.11, 1.92) but not in MSK (0.88). The strength differs between sources (Breslow–Day $p < 10^{-9}$), so pooled odds ratios are averages. Thin dark structures and ink show no association.

TABLE III
Artifact–melanoma association (training split). OR: odds ratio; MH: Mantel–Haenszel OR stratified by source.

| Artifact | Prev. | $P(\mathrm{m}\|\mathrm{a})$ | $P(\mathrm{m}\|\neg\mathrm{a})$ | OR | MH OR [95% CI] |
|---|---|---|---|---|---|
| Dark corners | 25.5% | .263 | .149 | 2.03 | 1.555 [1.43, 1.69] |
| Bright structures | 13.8% | .266 | .164 | 1.85 | 1.751 [1.59, 1.92] |
| Thin dark struct. | 24.2% | .174 | .179 | 0.97 | 0.988 [0.90, 1.08] |
| Ink / patches | 1.6% | .202 | .178 | 1.18 | 1.114 [0.85, 1.46] |
| Any | 48.8% | .210 | .148 | 1.53 | 1.265 [1.17, 1.36] |

### B. Classification

On the ISIC test set, Ours reaches the highest AUC among the prototype models ($0.865 \pm 0.006$, ensemble $0.884$; Table IV); its ensemble exceeds those of PPN-std, PPN and PPN-FA by 0.022, 0.018 and 0.017 (95% CI $[0.009, 0.036]$, $[0.004, 0.032]$, $[0.005, 0.031]$; Holm-adjusted $p = 0.012$, $0.042$, $0.029$), its seeds vary less (std 0.006 vs. 0.040 for PPN), and the three Ours seeds beat each of the nine baseline seeds (worst 0.8592 vs. best 0.8526; on Derm7pt-dermoscopic 0.7405 vs. 0.7343), a descriptive check; of seven Ours seeds (Section VI-F; $0.863\pm0.012$, range 0.842–0.880), one falls below the best baseline seed. At the validation threshold, Ours trades sensitivity for specificity. The black-box CNN remains more accurate (0.912 vs. 0.884, $p = 0.009$), a price of about 3 AUC points for interpretability (4.6 for PPN). Without the last-layer re-optimization (five-epoch last layer, epoch chosen on validation) all prototype models score lower, but Ours stays best ($0.846\pm0.007$; baselines 0.765–0.797, ablations 0.787–0.841), significantly so against the baselines ($p \leq 0.001$), and loses least (0.019 vs. 0.039–0.079 for them).

Externally, Ours has the highest AUC of the four main prototype models on Derm7pt-dermoscopic (0.749) and Derm7pt-clinical (0.657). On the dermoscopic images its ensemble beats PPN-FA and PPN-std ($p = 1.7 \times 10^{-5}$, $2.8 \times 10^{-6}$, also after Holm correction) but not PPN ($p = 0.20$, high seed variance); on the clinical images its advantage over PPN-std and PPN ($p = 0.023$, $0.046$) does not survive Holm correction ($p = 0.069$, $0.092$), and over PPN-FA it is not significant ($p = 0.30$). On PAD-UFES-20 (52 melanoma images of 36 lesions), Ours beats PPN-std ($p = 0.001$) and does not differ significantly from PPN and PPN-FA ($p \geq 0.23$); its mean AUC there (0.647, std 0.108) is even below that of PPN (0.665). The black-box EPVT reports 0.837 on Derm7pt-dermoscopic [7], but for melanoma versus benign, another split and a larger ViT-B/16 backbone; our CNN reaches 0.799.

### C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table V, Fig. 2(a)): in images with artifacts 23.1% of its hot region lies on them, 1.5 times their area share. The prototype

TABLE IV

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% CI; $p$: ENSEMBLE AGAINST OURS ON ISIC TEST (PAIRED BOOTSTRAP OVER LESION GROUPS). OURS: TWO CORE COMPONENTS AND THE AUXILIARY $\mathcal{L}_{\text{art}}$; OURS-NOART: THE TWO CORE COMPONENTS ALONE; PPN-FA: IAIA-BL LOSS ON $7 \times 7$ MAPS (ORIGINAL FORMULATIONS IN TABLE VIII).

| | ISIC 2019 test ($n$ = 2,320) | | | | | | External AUC | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | AUC | BAcc | Sens. | Spec. | Ens. AUC [95% CI] | $p$ | Derm7pt-derm | Derm7pt-clin | PAD-UFES-20 |
| CNN (black box) | .902±.006 | .814±.011 | .808±.044 | .820±.040 | .912 [.884, .935] | .009 | .799±.023 | .704±.022 | .721±.022 |
| PPN-std | .835±.013 | .752±.014 | .769±.063 | .734±.040 | .861 [.827, .888] | .002 | .677±.027 | .624±.023 | .587±.019 |
| PPN | .825±.040 | .758±.010 | **.786±.068** | .730±.087 | .866 [.830, .894] | .014 | .688±.072 | .618±.038 | **.665±.016** |
| PPN-FA (7×7) | .844±.008 | .744±.021 | .689±.045 | .800±.005 | .866 [.830, .895] | .007 | .696±.024 | .640±.013 | .630±.059 |
| **Ours** | **.865±.006** | **.780±.014** | .734±.075 | **.825±.046** | **.884 [.850, .910]** | – | **.749±.009** | **.657±.019** | .647±.108 |
| Ours-noLes | .830±.015 | .746±.030 | .766±.091 | .726±.032 | .856 [.821, .885] | <.001 | .717±.016 | .636±.033 | .628±.047 |
| Ours-noArt | .859±.011 | .772±.003 | .733±.056 | .811±.054 | .877 [.845, .904] | .22 | .752±.014 | .637±.030 | .678±.018 |
| Ours-stdPush | .863±.005 | .785±.002 | .823±.043 | .747±.047 | .884 [.852, .911] | .86 | .747±.005 | .670±.019 | .658±.108 |

TABLE V

WHERE THE MODELS LOOK (MEAN OVER 3 SEEDS). LES.: HOT-REGION SHARE ON THE LESION; ART.: SHARE ON ARTIFACTS IN IMAGES WITH ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; PT.: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES PROJECTED ONTO CELLS WITH ≥50% LESION (OF 20); DC–INK: SHARE ON THE GIVEN ARTIFACT IN IMAGES THAT CONTAIN IT; D7: DERM7PT-DERMOSCOPIC.

| | ISIC test | | | | | ISIC test, by artifact type | | | | D7 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Les.↑ | Art.↓ | Enr.↓ | Pt.↓ | L-pr. | DC | TS | SP | INK | Les. | Pt. | Enr. |
| CNN | .466 | .231 | 1.52 | .117 | – | .365 | .016 | .017 | .029 | .605 | .153 | .63 |
| PPN-std | .518 | .082 | 0.54 | .050 | 12.0 | .123 | **.010** | .015 | .049 | .749 | .075 | .25 |
| PPN | .562 | .066 | 0.46 | .042 | 14.3 | .098 | **.010** | .012 | .036 | **.789** | .044 | .24 |
| PPN-FA | .530 | .102 | 0.68 | .050 | 10.0 | .158 | **.010** | **.010** | .056 | .760 | .077 | .28 |
| **Ours** | **.589** | **.048** | **0.37** | **.013** | **20.0** | **.067** | .011 | .016 | **.032** | .734 | **.015** | **.18** |
| Ours-noLes | .565 | .106 | 0.67 | .053 | 20.0 | .162 | .011 | .013 | .044 | .772 | .057 | .30 |
| Ours-noArt | .594 | .042 | 0.40 | .018 | 20.0 | .056 | .012 | .020 | .040 | .713 | .017 | .23 |
| Ours-stdPush | .515 | .086 | 0.58 | .034 | 7.7 | .129 | .012 | .019 | .030 | .633 | .033 | .35 |

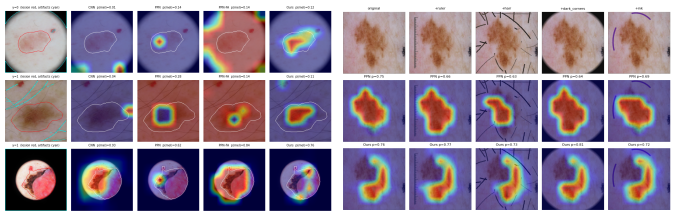


Fig. 2. (a, left) ISIC test images with artifacts (lesion red, detected artifacts cyan); CNN: class-activation map; prototype models: map of the most activated prototype; $p$: melanoma probability (seed 0). The CNN and PPN-FA respond to dark corners and borders, the top prototype of Ours stays on the lesion. (b, right) Counterfactual example with unchanged lesion pixels (not selected: the second of the first two test images that receive a ruler); rows 2–3: map of the prototype most active on the original image and $p$ for PPN and Ours; the map of Ours keeps its place, that of PPN shifts with added hair.

models look at artifacts less than their area suggests, Ours least (4.8±1.5% of the hot region, maximum on an artifact in 1.3±0.5% of images; ±: std over seeds), with the largest lesion fraction (58.9%). All 20 prototypes of every Ours run lie on lesion cells ($M^L \geq 0.5$; Fig. 3), whereas PPN-FA projects on average 10.0 onto lesion and 2.0 onto artifact cells. Most attention on artifacts falls on dark corners, where Ours has the smallest share (Table V). All differences between Ours and each other main model in lesion share, artifact share, enrichment and pointing are significant on the test sample (Wilcoxon on lesion-group means, $p \leq 5.6 \times 10^{-4}$; artifact shares in the primary family: Holm-adjusted $p \leq 2.2 \times 10^{-5}$); across training runs they overlap with PPN and PPN-std (Section VI-D). On Derm7pt-dermoscopic, Ours again has the lowest artifact share, enrichment and pointing rate ($p \leq 8.6 \times 10^{-3}$), but the three prototype baselines place more of their hot region on the lesion ($p \leq 1.2 \times 10^{-4}$; Table V).

**Receptive field of the prototypes.** The effective receptive field (ERF) of each prototype's source cell, the input-gradient magnitude of its latent vector on the source image (eight random sign projections), has only 8–12% of its mass inside the cell for every model, and half of it spreads over 8–11% of the image. For Ours, 69 ± 3% lies on the lesion (10th percentile over prototypes 42%), against 48–53% for PPN-std, PPN and PPN-FA (10th percentiles 11–21%), and 2.5% on detected artifacts (4.3%, 2.7%, 7.3%). This stems from where prototypes are taken, not from a narrower field: the constraint selects source images whose lesion covers 54% of the image (baselines 40%), and relative to the lesion area every model concentrates its ERF on the lesion by a median factor of only 1.2–1.3. Ours-stdPush drops to 47%, Ours-noLes and Ours-noArt keep 71–72%. For 13% of the prototypes of Ours (baselines 22–28%) the ERF draws less on the lesion than its area would: anchoring covers the projected cell and most, not all, of its receptive field (Fig. 3).

### D. Counterfactual robustness

Table VI summarizes the interventions (2,063–2,318 modified images per insertion type). Adding dark corners *increases* the melanoma probability of every model (mean signed change +0.041 CNN, +0.048 PPN, +0.020 Ours), matching the association in Table III: the models have absorbed the shortcut, Ours least. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers (7.5%→3.6%), reduces it for ink (9.5%→5.6%) and hair (13.9%→9.2%), and flips fewer decisions than the CNN for all four insertions. These reductions are significant against every other main model for all four insertions (Wilcoxon on lesion-group means, $p \leq 4.9 \times 10^{-8}$ against the prototype baselines, Holm-adjusted $p \leq 3.9 \times 10^{-7}$ for dark corners and rulers; $p \leq 0.008$ against the CNN), as is the smaller $|\Delta p|$ against PPN ($p < 10^{-27}$) and, except for hair ($p = 0.09$), the CNN ($p \leq 2.3 \times 10^{-4}$). Removing real hair

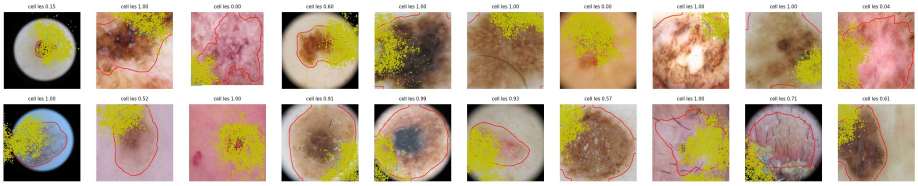


Fig. 3. Source images of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0, with the projected cell (cyan; its lesion fraction above each image), the lesion (red) and the smallest region holding half of the prototype's effective receptive field (yellow). Four PPN prototypes come from background cells (lesion fraction 0.00–0.15) and draw on skin and vignette; all of Ours come from lesion cells ($\geq 0.52$, no detected artifact) and draw mostly, though not only, on the lesion.

TABLE VI

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP: DECISION CHANGES (%); $|\Delta p|$: MEAN ABSOLUTE CHANGE OF THE MELANOMA PROBABILITY; $\rho$: SPEARMAN CORRELATION OF THE TOP-1 PROTOTYPE MAP BEFORE/AFTER; CAP.: SHARE OF THE NEW HOT REGION ON INSERTED PIXELS (%). THE NULL CONTROL IS NOT RANKED.

| | +Dark corners | | +Ruler | | +Hair | | +Ink | | Remove hair/ruler | | Null control | | Expl.: dark corners | | Expl.: ruler | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Flip | $\lvert\Delta p\rvert$ | Flip | $\lvert\Delta p\rvert$ | Flip | $\lvert\Delta p\rvert$ | Flip | $\lvert\Delta p\rvert$ | Flip | $\lvert\Delta p\rvert$ | Flip | $\lvert\Delta p\rvert$ | $\rho\uparrow$ | Cap.$\downarrow$ | $\rho\uparrow$ | Cap.$\downarrow$ |
| CNN | 6.1 | .049 | 5.6 | .049 | 12.5 | .104 | 9.7 | .080 | 4.7 | .036 | 1.0 | .008 | – | – | – | – |
| PPN-std | 10.9 | .060 | 8.3 | .041 | 14.9 | .092 | 9.9 | .058 | 8.8 | .036 | 4.3 | .013 | .635 | 7.1 | .773 | 1.5 |
| PPN | 10.9 | .067 | 7.5 | .045 | 13.9 | .096 | 9.5 | .062 | 7.4 | .045 | 3.6 | .019 | .680 | 6.1 | .804 | 1.1 |
| PPN-FA | 10.6 | .049 | 6.5 | .028 | 15.1 | .084 | 9.1 | .046 | 7.2 | **.030** | 3.2 | .010 | .714 | 11.4 | .874 | 1.1 |
| **Ours** | **5.2** | **.035** | **3.6** | **.025** | **9.2** | **.081** | **5.6** | **.045** | **4.4** | .032 | 1.6 | .011 | **.918** | **3.4** | **.952** | **0.7** |
| Ours-noLes | 10.2 | .053 | 8.7 | .040 | 14.3 | .083 | 10.1 | .049 | 8.3 | .042 | 3.9 | .016 | .705 | 8.6 | .827 | 1.7 |
| Ours-noArt | 5.4 | .036 | 3.5 | .023 | 8.7 | .061 | 6.1 | .041 | 4.4 | .027 | 1.0 | .011 | .884 | 3.4 | .949 | 1.0 |
| Ours-stdPush | 4.5 | .034 | 4.1 | .028 | 9.6 | .077 | 5.8 | .045 | 4.5 | .033 | 1.9 | .011 | .923 | 7.5 | .944 | 1.4 |

TABLE VII

REAL, UNEDITED ARTIFACTS ON BENIGN TEST LESIONS. $A$: PROBABILITY THAT A BENIGN LESION WITH THE ARTIFACT SCORES HIGHER THAN ONE WITHOUT IT (SEED ENSEMBLES; 0.5: NO EFFECT); FPR: FALSE-POSITIVE RATE (%) WITH/WITHOUT THE ARTIFACT (VALIDATION THRESHOLD, MEAN OVER SEEDS). BENIGN IMAGES WITH/WITHOUT: 447/1,483 (ISIC, DC), 251/1,679 (ISIC, SP), 515/244 (DERM7PT, DC).

| | ISIC, dark corners | | ISIC, bright struct. | | Derm7pt-clin., DC | |
|---|---|---|---|---|---|---|
| Model | $A\downarrow$ | FPR | $A\downarrow$ | FPR | $A\downarrow$ | FPR |
| CNN | .57 | 22/17 | .58 | 19/18 | .59 | 18/9 |
| PPN-std | .58 | 31/25 | .61 | 32/26 | .82 | 60/21 |
| PPN | .61 | 30/26 | .60 | 33/26 | .78 | 35/11 |
| PPN-FA | .66 | 33/16 | .62 | 26/19 | .80 | 46/15 |
| **Ours** | **.57** | **20/17** | **.59** | **19/17** | **.69** | **24/12** |
| Ours-noLes | .66 | 37/25 | .63 | 37/26 | .77 | 50/24 |
| Ours-noArt | .58 | 20/19 | .62 | 25/18 | .75 | 37/12 |
| Ours-stdPush | .57 | 28/25 | .59 | 29/25 | .65 | 33/19 |

and ruler marks changes more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so the sensitivity stems from the artifacts, not from inpainting.

Its explanations also move the least (Fig. 2(b)): after adding dark corners its top-1 map keeps $\rho = 0.92$ (PPN 0.68), and only 3.4% of its hot region (per seed 0.2–7.7%) moves onto the inserted pixels (PPN 6.1%, PPN-FA 11.4%; $p < 10^{-84}$). The exception is ink, drawn just outside the lesion border, where anchored prototypes respond: 1.8% of the hot region of Ours falls on it vs. 1.0% for PPN, though Ours still flips fewer decisions.

**Seed level.** Including the four further Ours networks of Section VI-F (seven seeds), every Ours seed flips fewer decisions under inserted dark corners and ink than every seed of PPN-std, PPN and PPN-FA (3.3–7.5% vs. 7.6–15.4%; 4.4–8.0% vs. 8.0–11.2%, i.e., at most 184 vs. at least 185 of 2,301 images) and keeps a more stable map than every seed of PPN-std and PPN-FA ($\rho$ 0.85–0.93 vs. $\leq$0.77); $\rho$ overlaps with one PPN seed (0.87) and the mask-loss variants ($\leq$0.90). The artifact share (3.0–8.9%) lies below every CNN seed ($\geq$20.7%) but overlaps with the prototype baselines and the Correia formulation, and the captured share overlaps for all models (Ours 0.2–7.7%, baselines 0.1–15.4%).

### *E. Shortcuts in unedited images*

As a complementary test without editing, we ask whether artifacts present in the test images raise the melanoma score of *benign* lesions, measured by the threshold-free probability $A$ of Table VII ($A = 0.5$: no effect). On ISIC, dark corners double the false-positive rate of PPN-FA (16%→33%) but raise that of Ours only from 17% to 20%. The effect $A$ of Ours (0.57) is significantly smaller than that of PPN, PPN-FA and Ours-noLes (0.61–0.66; $p < 0.001$) and not significantly different from that of PPN-std and the CNN ($p \geq 0.77$): our weak mask loss leaves the real dark-corner shortcut in place, lesion attraction keeps it small, and the original mask-loss formulations reduce it as well (Section VI-G). The same holds for bright structures, where Ours has the smallest effect of the prototype models (0.59 vs. 0.60–0.62; $p \leq 0.038$ against PPN-std and PPN-FA), and most clearly for dark border regions in the clinical Derm7pt photographs (0.69 vs. 0.78–0.82; DeLong $p \leq 1.4 \times 10^{-6}$). The effect of the CNN is not larger than that of Ours, although it attends to dark corners most.

### *F. Ablation of the two core components*

**Lesion attraction** is the term behind the accuracy gain: without it (Ours-noLes) the test AUC falls to 0.830 ($p <$

TABLE VIII

MASK-LOSS VARIANTS (ISIC TEST UNLESS NOTED; MEAN OVER 3 SEEDS). PPN-FA: $7 \times 7$ MAPS, BATCH MEAN; CORREIA [12], IAIA-BL [16]: $224 \times 224$ MAPS, BATCH SUM, $\lambda = 10^{-3}$. D7-D: DERM7PT-DERMOSCOPIC AUC; L-PR., ART.: AS IN TABLE V; FLIP, $\rho$: INSERTED DARK CORNERS; $A$: REAL DARK CORNERS; OURS+CORREIA: OURS PLUS THE CORREIA LOSS.

| Mask loss | AUC | D7-d | L-pr. | Art.↓ | Flip↓ | $\rho\uparrow$ | $A\downarrow$ |
|---|---|---|---|---|---|---|---|
| None (PPN) | .825±.040 | .688 | 14.3 | .066 | 10.9 | .68 | .61 |
| PPN-FA, $10^{-3}$ | .844±.008 | .696 | 10.0 | .102 | 10.6 | .71 | .66 |
| PPN-FA, $10^{-2}$ | .833±.018 | .695 | 16.0 | .121 | 12.2 | .80 | .66 |
| PPN-FA, $10^{-1}$ | .849±.006 | .711 | 9.3 | .061 | 7.2 | .84 | .60 |
| Correia | .867±.011 | **.771** | 13.0 | .141 | 6.5 | .87 | .57 |
| IAIA-BL | .845±.011 | .715 | 18.0 | **.039** | 6.2 | .88 | .58 |
| Ours (no mask loss) | .865±.006 | .749 | **20.0** | .048 | 5.2 | **.92** | .57 |
| Ours+Correia | **.871±.004** | .736 | **20.0** | .043 | **5.0** | .87 | **.53** |

TABLE IX

ROBUSTNESS TO MASK QUALITY (ISIC TEST, MEAN OVER 3 SEEDS): ATTRIBUTION AS IN TABLE V ON IMAGES WITH EXPERT (HAM10000, $n = 900$) AND PREDICTED MASKS (BCN20000, MSK, $n = 1{,}420$); LESION SHARE WITH ALL LESION MASKS ERODED ($-5$) OR DILATED ($+5$) BY 5 PX; AUC PER SOURCE ($n$/MELANOMAS: BCN 1,151/248, HAM 900/93, MSK 269/49).

| Model | Expert masks Les.↑ | Art.↓ | Pt.↓ | Predicted masks Les.↑ | Art.↓ | Pt.↓ | Lesion $-5$ | $+5$ | AUC by source BCN | HAM | MSK |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CNN | .444 | .043 | .015 | .480 | .277 | .182 | .434 | .496 | .883 | .934 | .861 |
| PPN-std | .416 | .016 | .004 | .582 | .098 | .079 | .480 | .552 | .825 | .880 | .711 |
| PPN | .446 | .014 | .005 | **.635** | .079 | .065 | .525 | .595 | .819 | .860 | .692 |
| PPN-FA | .423 | .019 | .004 | .599 | .123 | .080 | .491 | .567 | .834 | .896 | .699 |
| **Ours** | **.534** | **.013** | **.002** | .624 | **.056** | **.021** | **.534** | **.640** | **.852** | **.906** | **.761** |
| Ours-noLes | .503 | .023 | .007 | .603 | .126 | .082 | .525 | .601 | .815 | .879 | .704 |
| Ours-noArt | .543 | .013 | .003 | .626 | .050 | .028 | .541 | .644 | .853 | .893 | .737 |
| Ours-stdPush | .459 | .018 | .003 | .550 | .103 | .054 | .466 | .561 | .852 | .904 | .749 |

$0.001$), the artifact share of the hot region more than doubles (4.8%→10.6%), and the flip rates for dark corners, rulers and ink roughly double. Since Ours-noLes still uses the constrained projection, all of its prototypes lie on lesion cells: a correct projection alone is not enough, the attention during training has to be anchored as well.

**The constrained projection** determines what the prototypes *are* (Ours-stdPush: 7.7 of 20 on lesion cells) but changes neither the accuracy (ensemble difference within $[-0.006, 0.005]$, 95% CI) nor, significantly, the flip rates (dark corners 4.5% vs. 5.2%; $p \geq 0.13$): decision robustness comes from lesion attraction. Its map differences in Tables V and VI (artifact share 8.6% vs. 4.8%, captured share 7.5% vs. 3.4%) are not a systematic effect of the projection. We projected six independently trained networks both ways: seeds 0 and 2, where Ours and Ours-stdPush share one network (identical test latents; seed 1 diverged), and four new seeds whose trained prototypes we stored before the projection. In every network the constraint moved the prototypes onto lesion cells (6–11 $\rightarrow$ 20 of 20; exact Wilcoxon $p = 0.031$, the smallest possible with six pairs), but it changed neither the AUC (mean difference $+0.0001$) nor, consistently, the flip rates, artifact share, captured share or $\rho$ (differences of both signs, $p \geq 0.22$).

**Projection thresholds.** On the four new networks, projected from their stored trained prototypes, lesion thresholds of 0.3–0.9 (artifact fraction $\leq$0.05) and artifact limits of 0, 0.2 or none (lesion fraction $\geq$0.5) put 14.8–20 of 20 prototypes on lesion cells (mean lesion fraction of the source cells 0.78–0.99 vs. 0.47 without constraint), whereas the AUC (0.861–0.863), dark-corner flips (5.8–6.1%), artifact share (5.1–5.5%) and $\rho$ (0.885–0.886) stay at the standard projection's values (0.861, 5.8%, 5.2%, 0.886); only the strictest threshold (0.9) removes most border cells (2.3 vs. 8.5 at 0.5).

### *G. Strength of the mask-loss baseline*

IAIA-BL [16] and Correia *et al.* [12] compute the fine-annotation loss at $224 \times 224$ and sum it over the batch, which makes the same $\lambda = 10^{-3}$ about three orders of magnitude stronger than in PPN-FA (factors 32 and 64; Correia *et al.* penalize own-class prototypes only). We therefore trained both original formulations and PPN-FA with $\lambda = 10^{-2}$ and $10^{-1}$, three seeds each (Table VIII; secondary tests). The ISIC AUC of the Correia formulation does not differ significantly from that of Ours (ensemble difference $-0.001$, 95% CI $[-0.012, 0.011]$), and it generalizes better (Derm7pt-dermoscopic 0.771 vs. 0.749, Derm7pt-clinical 0.702 vs. 0.657, PAD-UFES-20 0.699 vs. 0.647; $p \leq 0.028$), yet only 13 of its 20 prototypes are lesion cells and its hot region lies three times more on artifacts (14.1% vs. 4.8%, $p < 10^{-29}$; seeds 8.1–24.6% vs. 3.0–8.9%). In the primary family, Ours remains significantly better than this formulation on artifact share and dark-corner flips (Holm $p < 10^{-28}$, $p = 0.029$), not on AUC or ruler flips. The IAIA-BL formulation keeps its hot region on the lesion (83%) and off artifacts (3.9%) better than Ours ($p < 0.002$) but is less accurate (0.845; ensembles $p = 0.057$) and leaves 2 of 20 prototypes off lesion cells. Every variant flips more decisions than Ours under inserted dark corners (6.2–12.2% vs. 5.2%) and ink (6.6–10.7% vs. 5.6%; $p \leq 0.02$; per seed, the original formulations, and for dark corners also $\lambda = 10^{-1}$, overlap with Ours); for rulers only the original formulations ($p \geq 0.052$), and for hair only the Correia one ($p = 0.37$), do not differ significantly from Ours, and none keeps its explanation as stable (Section VI-D). The real dark-corner effect of both original formulations does not differ significantly from that of Ours ($A$ 0.57–0.58 vs. 0.57, $p \geq 0.26$), unlike our weak version (0.66). Mask supervision thus shapes where the network looks and helps it generalize, while anchoring fixes what the prototypes are and keeps decisions stable: the two are complementary.

**Combination.** Adding the Correia loss to our model (Ours+Correia) gives the highest ISIC AUC of all prototype models (0.871±0.004; $p \geq 0.32$ against Ours and Correia), more lesion-focused maps than Ours (lesion share 73% vs. 59%, seeds 59–85% vs. 49–69%; artifact share 4.3% vs. 4.8%, captured share 2.2% vs. 3.4%; $p < 10^{-7}$), the smallest real dark-corner effect of all models ($A = 0.53$; $p < 0.001$ against both) and similar flip rates ($p \geq 0.26$; hair 7.9% vs. 9.2%, $p = 0.02$). It does not inherit the Derm7pt advantage of the mask loss (0.736 and 0.675 vs. 0.771 and 0.702; DeLong $p \leq 0.016$) but has the highest PAD-UFES-20 AUC of all

models (0.737, CNN 0.721; $p \leq 0.001$ against both), and its maps are less stable than those of Ours ($\rho$ 0.87 vs. 0.92).

### H. Artifact suppression: a largely negative result

The auxiliary term penalizes attention on artifacts directly; with $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$, the attention mass of prototype $j$ on the artifacts of image $i$,

$$\mathcal{L}_{\text{art}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}. \tag{3}$$

Removing it (Ours-noArt) changes the accuracy significantly neither on ISIC (ensemble difference 0.007, 95% CI $[-0.004, 0.018]$) nor on any external set ($p \geq 0.058$); several counterfactual measures are even better without it (e.g., $|\Delta p|$ 0.061 vs. 0.081 for added hair, $p < 10^{-40}$), and the attribution changes little (artifact share 4.2% vs. 4.8%, $p = 0.36$; pointing 1.8% vs. 1.3%, $p = 5.7 \times 10^{-5}$). The two core components alone carry the explanation gains: against PPN-std, PPN and PPN-FA, Ours-noArt has more hot region on the lesion and less on artifacts and flips fewer decisions under all four insertions ($p \leq 5.2 \times 10^{-4}$), but its AUC gain over them (0.010–0.016) is not significant with lesion groups resampled ($p = 0.073$–$0.17$; image-level DeLong $p = 0.005$ against PPN-std), so the significant accuracy gain belongs to the main model. $\mathcal{L}_{\text{art}}$ mainly weakens the response to real bright structures and to dark border regions in clinical images ($A$ 0.59 vs. 0.62 and 0.69 vs. 0.75; $p < 0.001$, Table VII); otherwise its effect is small. We therefore keep $\mathcal{L}_{\text{art}}$ as an auxiliary term but do not claim it as a contribution.

### I. Robustness to mask quality

Predicted masks could bias the results if a model looked "on the lesion" only by the measure of an imperfect mask. We therefore split the ISIC test set by mask provenance and perturb the lesion masks (Table IX).

**Expert masks alone.** On the HAM10000 test images, Ours places 53.4% of its hot region on the lesion (prototype baselines 41.6–44.6%, CNN 44.4%), a *larger* advantage than on the whole test set (+8.8 vs. +2.7 points over the best baseline), so it is not produced by errors of the predicted masks; its artifact scores are the lowest there too. On the predicted-mask images Ours has the lowest artifact share (5.6% vs. 7.9–27.7%), although PPN places slightly more on the lesion (63.5±1.5% vs. 62.4±2.6%).

**Perturbed masks.** Eroding or dilating every lesion mask by 5 pixels shifts the lesion fractions of all models alike and keeps the order of the five main models (Ours first; Table IX); artifact share, pointing and enrichment change by at most 0.011. The provenance guarantee itself depends on the mask border: with masks eroded by 5 pixels, on average 17 of the 20 prototypes of Ours remain on cells with ≥50% lesion (lowest 0.36), all 20 with dilated masks. **Classification per source.** Ours is the most accurate of the four main prototype models in every source, with expert training masks (HAM10000) and with predicted ones (BCN20000, MSK; Table IX).

**Counterfactuals per provenance.** A too small predicted mask could let inserted artifacts touch the true lesion, but on the expert-mask subset, where the protected region is exact, Ours flips the fewest decisions (dark corners 4.4% vs. 7.1% for the CNN and 10.0–10.3% for the prototype baselines; rulers 2.3% vs. 3.9% and 4.7–6.1%), and on the predicted-mask subset the fewest among the prototype models (dark corners 5.7% vs. 10.7–11.5%, CNN 5.5%; rulers 4.4% vs. 7.7–9.7%, CNN 6.7%), with the smallest $|\Delta p|$ in both. The conclusions thus hold on expert masks alone, under ±5-pixel mask errors and in every source.

## VII. Discussion and Limitations

The two components play different roles: lesion attraction changes what the network learns and makes it both more accurate and less dependent on artifacts, whereas the constrained projection changes what it shows to the user: it guarantees the provenance of each explanation, which matters for presentation and audit, but is not itself a mechanism against shortcuts. The evidence thus supports claims about prototype provenance and robustness to artifacts, not about accuracy: our AUC does not differ significantly from that of the strongest mask-supervised ProtoPNet and stays below that of the black-box CNN. Lesion masks are needed only in training, come from public expert segmentations of one source (no new annotation for a new dermoscopic dataset), and enter only as coarse $7 \times 7$ cell fractions, so our conclusions tolerate boundary errors (Section VI-I). Anchoring keeps the lesion border: 9.3 of 20 prototypes of Ours are border cells (lesion fraction 0.5–0.95; baselines and mask-loss variants 1.7–5.0), which is why ink just outside the border draws some of its attention (Section VI-D).

**Limitations.** (i) A gap of about 3 AUC points to the black-box CNN remains; Correia *et al.* [12] found prototype models on par with CNNs, but on ISIC validation data and after choosing the best of many configurations. (ii) Predicted BCN20000 and MSK masks are validated only indirectly (held-out HAM10000 images, PH2, subset analysis), and the artifact detectors miss many artifacts (Section IV), so our evidence concerns detected and inserted artifacts, not every artifact in real images. (iii) Inserted artifacts are synthetic and may lie outside the training distribution; the removal of real artifacts, a null control and the test on unedited images mitigate this, but on unedited images the advantage of Ours is significant only against PPN, PPN-FA and Ours-noLes, and these effects may be partly confounded by lesion appearance. (iv) The $7 \times 7$ grid limits the spatial resolution of the explanations, and class-activation and prototype maps differ in kind, so attribution comparisons between the CNN and the prototype models are only indicative. (v) External gains over the prototype baselines are only partly significant (Section VI-B), and the original mask loss generalizes better (Section VI-G). (vi) One backbone and one prototype architecture were studied, as a controlled comparison requires, although both components apply to any prototype network with a projection step; the loss weights were fixed a priori;

and training variability rests on three seeds per configuration (seven for Ours, six paired networks for the projection), between which explanation metrics vary strongly (Section VI-D). (vii) No dermatologist rated the prototypes, and this retrospective study does not establish clinical readiness, which would require prospective validation in other populations.

## VIII. Conclusion

Classifiers trained on ISIC 2019 exploit artifact shortcuts. In our same-backbone, leakage-controlled comparison, a prototype network anchored to the lesion without new annotation, by lesion attraction on the prototype attention and a projection restricted to lesion-dominated cells, reached an ISIC AUC not significantly different from that of the strongest mask-supervised ProtoPNet, kept all prototypes on lesion-dominated cells by design, and, under artifacts inserted around an unchanged lesion, flipped significantly fewer decisions than every baseline for dark corners and ink, never significantly more for any insertion, with more stable explanations, and in each of seven seeds fewer dark-corner and ink flips than any seed of the ProtoPNet baselines; lesion attraction drove accuracy and robustness, the projection fixed provenance without changing either, and the auxiliary artifact-suppression term had only small, mixed effects. Adding the original mask loss further weakened the real dark-corner shortcut, while that loss alone generalized better to Derm7pt. **Future work** will test lesion anchoring in other prototype architectures (e.g., PIP-Net [13]) and backbones, seek a combination with mask supervision that also keeps its better external generalization, replace the conservative artifact detectors by learned ones with higher recall, and evaluate the prototypes with dermatologists in a reader study; the counterfactual protocol can check other interpretable classifiers as well.

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